Manifold for Type 1 Diabetes – Restoration

University of British Columbia — stem-cell-derived β-cell differentiation, maturation, and graft-performance analytics

1 · THE CLINICAL PROBLEM

The translation of stem-cell-derived β-cells from in-vitro protocol to insulin-independent clinical outcome is now empirically established — VX-880's initial cohort being the proof-of-principle. What is not yet established, at the analytic level required for efficient next-generation program design, is the directed causal structure connecting differentiation-protocol variables, maturation-state biomarkers, and post-graft functional outcome. The UBC Life Sciences Institute — with direct lineage through the ViaCyte program and continuing leadership in SC-β biology — holds a substrate unmatched elsewhere:

- **Differentiation-protocol variability.** Stage-specific signaling modulator concentrations, timing, and sequence produce SC-β populations with divergent maturation states, insulin-secretory dynamics, and post-transplant performance. The mapping from protocol parameters to functional outcome is currently characterized empirically; recovering the directed structure is the analytic tractability frontier.
- **Preclinical-to-clinical translation signal.** UBC's historical and continuing work bridging rodent and non-human primate preclinical outcomes with human clinical performance — including ViaCyte legacy data — provides the substrate for translation-prediction inference that newer labs cannot replicate.
- **Encapsulation, immune-evasion, and genetic engineering.** Gene-edited (including CRISPR-modified) and encapsulated SC-β architectures introduce new axes on the design space. Their ordering relative to conventional immunosuppression-based approaches requires causal structure on the joint outcome graph.

Three capabilities remain underdeveloped in current analytic practice across SC-β research:

- Unified causal modeling of differentiation-protocol parameters, maturation-state biomarkers, and post-graft functional outcome — recovering the directed structure from the joint data rather than pursuing marginal parameter-sweeps.
- Translation-prediction inference bridging rodent, non-human primate, and human performance — heterogeneous-species-effect learners calibrated on the legacy translation record.
- Principled prioritization of encapsulation, gene-editing, and immune-evasion variant programs on a unified causal graph, ranking modifications by attenuation of the dominant failure-mode cascade at each translation stage.

> Stem-cell-derived β-cells have crossed from biological possibility to clinical insulin-independence. The field's next five years will be defined by protocol-to-performance generalization and by next-generation architecture design — encapsulation, gene-editing, immune-evasion — which is fundamentally a causal-inference problem on the legacy translation record and the high-dimensional design space.

2 · WHAT MANIFOLD BRINGS

Manifold is a causal-intelligence platform for detecting structural fragility and regime transitions in complex multi-signal systems. Three engine capabilities form the basis of the UBC-configured deployment:

1. **Causal discovery (Spirtes / PC algorithm) on the differentiation-to-outcome record.** Recovers the directed structure among protocol parameters, maturation biomarkers, preclinical graft performance, and — where available — clinical outcome, providing the substrate for protocol-to-performance inference and next-generation design.
2. **Criticality estimators feeding a composite fragility score (ΩF).** Configured for preclinical-to-clinical translation — Gaussian process regression on dose-protocol response surfaces, heterogeneous-treatment-effect learners on genetic-edit variants, Bayesian change-point on maturation trajectories, species-effect models for translation prediction — with the composite score as a stable integration layer.
3. **Pearl do-calculus and interdiction on the design graph.** Encapsulation, gene-editing, and immune-evasion variant prioritization is framed as maximum-attenuation intervention on the inferred failure-mode cascade, producing principled rankings of candidate modifications rather than empirical program-by-program iteration.

3 · ΩF PILLAR MAPPING – SC-β DESIGN AND GRAFT-PERFORMANCE PHYSIOLOGY

ΩF PILLAR	SC-β READING
I — Interaction	Differentiation-protocol parameter × maturation-state × genetic-modification × graft-architecture interactions; SC-β × host-immune compatibility
R — Reserve	Maturation-state functional mass proxied by GSIS, insulin content, and expression-panel markers; post-graft functional mass across species
J — Junction	Immune–metabolic coupling across preclinical and clinical translation; encapsulation-barrier × host-immune coupling
C — Cascade	β-cell differentiation-to-maturation cascade (NKX6.1, MAFA, PDX1); dedifferentiation cascades under post-graft stress; encapsulation-fibrosis cascades
T — Temporal	Hazard of maturation-state regression, graft failure, or response loss given trajectory; time-to-functional-maturity; time-to-insulin-independence across species

4 · CRITICALITY ESTIMATOR CONFIGURATION

SC-β research data is high-dimensional across differentiation protocol and genetic-modification space, longitudinally rich per preparation, and cross-species in translation signal. The criticality layer is configured accordingly:

ESTIMATOR	WHAT IT CATCHES	PILLAR FEED
Gaussian process regression on protocol-parameter × outcome surfaces	Sparse dose-protocol surface estimation with calibrated uncertainty; prediction of untested protocol variants	I, R
Heterogeneous-treatment-effect learners (Causal Forest, BART) on genetic-modification variants	Gene-edit and encapsulation variant response differentiation at realistic preclinical sample sizes	I, C
Bayesian online change-point (BOCPD) on maturation-marker trajectories	Differentiation-state transition and dedifferentiation-onset detection with calibrated uncertainty	C, T
Cross-species translation models (hierarchical with species-effect)	Preclinical-to-clinical prediction bridging rodent, non-human primate, and human data	R, T

Additional estimators available within the same layer for subsequent phases:

- **Transfer entropy** across the differentiation-marker cascade — quantifies the Cascade-pillar directional coupling during SC-β maturation and dedifferentiation.
- **Persistent homology** on longitudinal single-cell maturation trajectories — topological regime detection across differentiation-state space, valuable for identifying maturation-arrested populations before functional readouts confirm.

5 · PROPOSED INITIAL ENGAGEMENT (PHASE 1)

The scope of the proposed Phase 1 engagement is defined to produce a deliverable of demonstrable analytic value to the UBC Life Sciences Institute within a contained time and data footprint.

- **Primary dataset.** UBC-resident differentiation-protocol and preclinical SC-β performance data, with per-protocol maturation-state biomarkers, graft-functional readouts (GSIS, C-peptide, insulin content), and post-transplant outcome across available species. Where accessible under existing collaboration frameworks, ViaCyte-legacy preclinical-to-clinical translation record as complementary input.
- **Access path.** UBC-internal Data Use Agreement, under the sponsorship of a Life Sciences Institute principal investigator; collaboration-framework access to legacy translation data where appropriate.
- **Deliverable.** Causal-graph characterization of the differentiation-to-post-graft-outcome pathway. Per-preparation **Ω**F trajectory integrating Gaussian-process, heterogeneous-treatment-effect, and change-point estimators. Pearl-do-calculus ranking of next-generation encapsulation and gene-editing modifications by attenuation of the dominant failure-mode cascade.
- **Success criteria.** Protocol-to-outcome causal graph consistent with pre-specified biological priors; preclinical-to-clinical translation prediction with calibration exceeding the Lab's current empirical baseline; interdiction-engine next-generation-variant rankings interpretable and actionable by UBC investigators.
- **Timeline.** Twelve to fourteen weeks from data access.
- **Excluded from Phase 1** (reserved for the Phase 2 translational arm): prospective human-study design, live clinical-trial data ingestion, bespoke assay development.

6 · RATIONALE AND TIMING

- **Replacement has crossed the proof-of-principle threshold.** Insulin-independence in early-phase Vertex VX-880 patients is a stage-change for the field. The next five years will be defined by protocol-to-performance generalization and by next-generation architecture design — both analytic problems before they are biological ones.
- **UBC's translation lineage is an unusual substrate.** The combination of current SC-β biology leadership with the ViaCyte-era preclinical-to-clinical translation record is a dataset no newer lab can reproduce. The analytic tools appropriate to cross-species translation inference have matured to production grade against this substrate.
- **Encapsulation and gene-edited architectures are the next design axes.** The acquisition-consolidated program landscape produces many parallel encapsulation, immune-evasion, and gene-edit variants; principled ordering of them requires causal structure on the joint outcome graph, which is Manifold's scope.
- **Relevant funding mechanisms are active.** The Breakthrough T1D Cures program, the Helmsley Charitable Trust Restoring Insulin Production stream, Canadian Institutes of Health Research, and NIH NIDDK all have active Requests for Proposals spanning SC-β analytics.
- **The platform is production-ready.** Manifold's causal, criticality, and interdiction engines are production-grade; configuration for the SC-β differentiation and graft-performance domains represents bounded, well-scoped work.

7 · PROPOSED PARTNERSHIP

Partnership with a principal investigator at the UBC Life Sciences Institute is sought for:

1. Retrospective differentiation-protocol and preclinical SC-β performance data access under UBC's existing Data Use Agreement.
2. Co-authorship on the protocol-to-performance causal-integration and next-generation-variant interdiction benchmark manuscript.
3. Co-application to Breakthrough T1D Cures, Helmsley Restoring Insulin Production, CIHR, or NIH NIDDK for the Phase 2 translational arm.

In exchange, Apex Analytica contributes the causal-inference engine, the **Ω**F framework, and the engineering team. Apex retains commercial rights to the platform; research intellectual property from the joint work is shared on standard academic–industry terms.